

Standards Coverage Matrix

Lab 1 Curriculum and Framework Alignment

Agent Failure Labs University Pilot Pack

Document Purpose

This matrix maps Agent Failure Lab 1 to selected cybersecurity and AI risk frameworks. The purpose is to help instructors, curriculum reviewers, and pilot evaluators understand which concepts the lab supports. This document does not claim that one lab fully satisfies any complete curriculum framework or professional standard.

1. Scope and Interpretation

Lab 1, *OpsMail Assistant: Indirect Prompt Injection via Poisoned Inbox*, is a focused practical exercise. It supports selected cybersecurity and AI security learning outcomes, especially those related to prompt injection, trust boundaries, sensitive-data disclosure, secure design, and trace-backed analysis.

The mappings below should be interpreted conservatively:

- **Direct** means the lab directly teaches or assesses the concept through hands-on work.
- **Supported** means the lab reinforces the concept but does not fully assess it.
- **Introduced** means the lab gives students initial exposure to the concept.
- **Not primary** means the concept is adjacent but not central to the lab.

2. Reference Frameworks

This matrix uses the following frameworks as curriculum and risk-alignment anchors:

- ACM/IEEE/AAAI Computer Science Curricula 2023, especially the Security knowledge area.
- ACM/IEEE/AIS SIGSEC/IFIP Cybersecurity Curricula 2017, especially broad cybersecurity knowledge areas.
- NIST NICE Workforce Framework for Cybersecurity.
- OWASP LLM and agentic application security concepts.
- NIST AI Risk Management Framework functions.
- MITRE ATLAS-style adversarial AI concepts.

Conservative Mapping Note

This document avoids overclaiming. Lab 1 is best understood as a standards-aligned practical module, not as a complete replacement for a full cybersecurity or AI security curriculum.

3. Lab 1 Concept Inventory

The following internal concept identifiers are used throughout the matrix.

ID	Concept
C1	Indirect prompt injection
C2	AI agent tool use
C3	Untrusted external content
C4	Trust-boundary failure
C5	Instruction/data separation
C6	Sensitive-data disclosure
C7	Least-privilege access
C8	Output policy enforcement
C9	Provenance-aware handling of external content
C10	Trace-backed evidence and event analysis
C11	Professional security reporting
C12	Mitigation design and residual-risk analysis
C13	Controlled cyber range rules of engagement
C14	Privacy and synthetic PII handling

4. Summary Coverage Matrix

Framework Area	Coverage	Lab 1 Alignment
Prompt injection / LLM application security	Direct	Students exploit indirect prompt injection through attacker-controlled email.
Secure software design	Direct	Students explain instruction/data separation and least-privilege failures.
Privacy and sensitive-data protection	Direct	Lab demonstrates synthetic PII disclosure.
Threat modeling	Direct	Students identify attacker capability, target asset, trust boundary, and success condition.
Logging, monitoring, and trace analysis	Direct	Students use event timelines and traces as evidence.
Risk management and mitigation	Supported	Students propose controls and residual-risk considerations.
Governance and compliance	Supported	Lab can support discussion of policy, audit framing, and disclosure rules.
Network security	Not primary	The lab does not focus on network protocols or packet-level security.
Cryptography	Not primary	The lab does not teach cryptographic primitives or protocol design.
Malware analysis	Not primary	The lab does not involve malware reverse engineering.

5. ACM/IEEE/AAAI CS2023 Security Mapping

The CS2023 Security knowledge area emphasizes a security mindset and secure treatment of systems, privacy, and software behavior. Lab 1 supports selected concepts within that area by giving students an applied agent-security exercise.

CS2023-Aligned Concept	Coverage	Agent Failure Concepts	Evidence in Lab
Security mindset	Direct	C3, C4, C5, C12	Students reason about adversarial use of ordinary email workflows.
Threats and attacks	Direct	C1, C3, C4	Students execute indirect prompt injection.
Privacy and sensitive information	Direct	C6, C8, C14	Students analyze synthetic home-address disclosure.
Secure software design	Direct	C5, C7, C8, C12	Students propose least privilege, output controls, and instruction/data separation.
Defensive mechanisms	Supported	C7, C8, C9, C12	Students recommend layered mitigations.
Professional and ethical practice	Supported	C11, C13	Students operate within rules of engagement and submit a professional report.
System monitoring and evidence	Supported	C10, C11	Students use trace and event evidence to support claims.

6. ACM CSEC 2017 Cybersecurity Mapping

CSEC 2017 organizes cybersecurity education around broad knowledge areas. Lab 1 aligns most strongly with software security, data security, system security, and human/organizational dimensions of cybersecurity.

CSEC Area	Knowledge	Coverage	Lab 1 Alignment
Data Security		Direct	The lab demonstrates unauthorized disclosure of synthetic sensitive data and asks students to propose disclosure controls.

Software Security	Direct	Students analyze an insecure AI application workflow and propose secure design changes.
System Security	Supported	The lab shows unsafe interaction between an agent runtime, tools, inbox data, and sensitive context.
Human Security	Supported	The exploit uses business-process framing and human-readable email content to influence agent behavior.
Organizational Security	Supported	Students discuss policy, audit framing, governance, and operational controls.
Societal Security	Introduced	The lab can motivate broader discussion of privacy and responsible deployment of AI assistants.
Connection / Network Security	Not primary	The lab does not focus on network-layer attack or defense.

7. NIST NICE Framework Mapping

The NICE Framework provides a common language for cybersecurity work, roles, and learner capabilities. Lab 1 maps most naturally to work involving secure design, protection and defense, investigation, and governance.

NICE Work Role Category	Coverage	Lab 1 Alignment
Design and Development	Direct	Students identify design flaws in an AI agent application and propose secure design changes.
Protection and Defense	Supported	Students analyze how security controls could prevent, detect, or limit disclosure.
Oversight and Governance	Supported	Students discuss privacy, data handling, policy controls, and residual risk.
Investigation	Supported	Students reconstruct the exploit chain using trace and event evidence.
Implementation and Operation	Introduced	Students observe runtime and tool behavior, but do not operate production infrastructure.
Research	Introduced	Students analyze an emerging AI security failure pattern.
Cyberspace Effects	Not primary	The lab does not involve offensive operations beyond the controlled educational environment.

7.1 NICE Capability Alignment

Learner Capability	Coverage	Evidence in Lab
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Identify vulnerabilities in software behavior	Direct	Student identifies indirect prompt injection and trust-boundary failure.
Analyze attacker capability and target assets	Direct	Student defines attacker access, target data, and success condition.
Interpret logs or event traces	Direct	Student uses trace stream and event timeline evidence.
Recommend mitigation strategies	Direct	Student proposes least privilege, output filtering, and instruction/data separation.
Communicate technical findings	Direct	Student submits a structured security report.
Operate production security systems	Not primary	The student does not administer real enterprise infrastructure.

8. OWASP LLM and Agentic Security Mapping

Lab 1 is strongly aligned with LLM application security and emerging agentic application security concepts.

OWASP-Aligned Area	Coverage	Lab 1 Alignment
Prompt Injection	Direct	Malicious instructions are embedded in an external email and later processed by the assistant.
Indirect Prompt Injection	Direct	The attacker influences the agent through inbox content rather than direct chat.
Sensitive Information Disclosure	Direct	The assistant reveals synthetic protected personal information.
Excessive Agency / Tool Misuse	Supported	The assistant's email-processing workflow allows untrusted content to influence behavior.
Instruction Hierarchy Failure	Supported	The lab demonstrates confusion between trusted instructions and external email content.
Insecure Output Handling	Supported	The final response is not blocked despite containing protected data.

9. NIST AI Risk Management Framework Mapping

The NIST AI RMF functions provide useful language for connecting the lab to AI governance and risk management.

AI RMF Function	Coverage	Lab 1 Alignment
Map	Direct	Students identify the context of use, target asset, attacker capability, trust boundary, and security objective.

Measure	Supported	Students observe and evaluate agent behavior using trace evidence and success criteria.
Manage	Direct	Students propose mitigations, controls, and residual-risk considerations.
Govern	Supported	Students discuss policy, disclosure rules, access control, and safe deployment practices.

10. MITRE ATLAS-Style Adversarial AI Mapping

Lab 1 can be described using adversarial AI concepts commonly associated with prompt injection, adversarial input, and exfiltration-style outcomes.

Adversarial AI Concept	Coverage	Lab 1 Alignment
Adversarial input to AI system	Direct	The malicious email is an attacker-controlled input processed by the AI assistant.
Prompt injection-style manipulation	Direct	The payload attempts to alter assistant behavior through natural-language instructions.
Indirect attack path	Direct	The attacker does not need to interact directly with the victim chat session.
Exfiltration-style outcome	Direct	The assistant reveals synthetic protected information.
Policy bypass framing	Supported	The payload may frame disclosure as a compliance or audit task.
Post-compromise operations	Not primary	The lab does not involve persistence, lateral movement, or real infrastructure compromise.

11. Assessment Evidence Mapping

The following table connects standards-aligned concepts to observable student evidence.

Learning Concept	Observable Evidence	Assessment Artifact
Indirect prompt injection	Malicious email payload and victim trigger prompt	Student report methodology section
Trust-boundary failure	Trace showing email content entering agent workflow	Trace evidence section
Sensitive-data disclosure	Final assistant response containing protected target information	Results section
Threat modeling	Attacker capability, limitation, target asset, and success condition	Threat model section

Secure design	Least privilege, instruction/data separation, output controls	Mitigation section
Trace analysis	Event timeline and tool-call interpretation	Evidence and results section
Professional reporting	Structured, concise technical explanation	Full report submission

12. Limitations and Non-Coverage

Lab 1 is intentionally focused. It does not cover every cybersecurity or AI security topic.

- It does not teach cryptographic protocol design.
- It does not teach malware analysis.
- It does not teach network packet analysis.
- It does not teach secure coding in a specific programming language.
- It does not teach cloud infrastructure hardening in depth.
- It does not teach training-time attacks such as data poisoning.
- It does not fully satisfy any entire curriculum framework by itself.

Instead, Lab 1 provides a focused practical module on AI agent security, indirect prompt injection, trace-backed analysis, and mitigation design.

13. Summary

Agent Failure Lab 1 provides direct, practical coverage of several important modern cybersecurity learning outcomes:

- prompt injection and LLM application security;
- trust-boundary analysis;
- sensitive-data disclosure;
- AI agent tool-use risk;
- secure design and least privilege;
- trace-backed evidence collection;
- professional security reporting.

The lab is strongest when used as a standards-aligned practical exercise inside a broader cyber-security, secure software engineering, privacy, or AI security course.

Recommended Standards Positioning

Agent Failure Lab 1 should be described as a focused, standards-aligned practical module. It supports selected curriculum and workforce concepts, but it should not be presented as fully satisfying any complete framework on its own.

14. End of Standards Coverage Matrix